

To Install or Not to Install: Why Businesses Choose On-Site Renewable Energy

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National Environmental Partnership Summit

May 11, 2006



Background

- Renewable energy lead for office
- Concern about low % of manufacturers applying for state grants
- Driving forces and common ground
- Tool to identify good candidates



Information Sources

- Industry organizations
- State funding organizations
- Businesses with on-site renewables
- Industry currently in the installation process



Types of Businesses

- Food processors
- Plastics
- Pharmaceuticals
- Aluminum extrusion
- Semiconductors
- Inverter manufacturer
- Beauty salon



Main Concerns for Those That Install

COST/BENEFIT

ENVIRONMENTAL



Cost/Benefit

- Paybacks typically 5-10 years
- Certainty about future energy prices
- Large energy users minimize or cut costs while growing business
- Stay competitive



Environmental

- Reliance on fossil fuels
- GHG emissions
- Concern for employees
- Positive affect on regional air quality



Technology-Specific Concerns

- Availability of resource
- Structural/roof issues
- Efficiency of technology



On-site Renewable Energy is Not for Everyone

- Struggling businesses
- Low energy prices
- No energy efficiency effort

Upper-level commitment is important



How to Choose the Right Renewable

- Availability of resource
- Energy demand of business
- Community acceptance



Good Solar Hot Water Candidates

- Have a relatively large demand for hot water
 - Beauty salons
 - Laundries
 - Food processing
 - Plating
 - Textiles
 - Pulp & paper
- Year around direct sun from 9am-4pm
- Space for installation
- Structurally sound location



Good Biomass Candidates

- Proximity to resource
- Demand for heat
- Sufficient space



Good PV Candidates

- Preferred installation area:
flat or SW-SE facing,
structurally sound, low-cost, minimal obstructions or shading
- If roof PV, where roof will not require replacement soon
- Where surcharges for peak electricity exist



Good Wind Candidates

- 24/7 operations
- >1000ft from nearest residence
- Class 3 or better wind (6.5m/s at hub height)
- On hill tops
- Away from trees, airports, and sandy soil



Fresh Hair - Solar Thermal



- Full service beauty salon
- Considered PV, but too expensive
- Projected 8yr payback
- \$9,400 50-60MBtu/yr project, ~3/4 of need
- Tax incentives pay 40%, no grant money
- Pro: Investment in company and environment
- Con: Water out for 1-2 days during installation and a couple hours during annual maintenance



Harbec Plastics - Wind



- Injection molder
- Considered solar, but better wind resource
- Projected 8-9yr payback
- \$375K 250kW project, no grants or tax benefits
- Pro: Everyone knows Harbec is environmentally responsible even if they don't know what Harbec does
- Con: Took 13 months to get planning board approval



Bixby - PV



bixby international corporation

- Plastic sheet extruder
- Considered wind, but initial investment too high
- Projected 6-7yr payback
- \$345K 51kW project, \$257K in tax incentives and state grants
- Pro: A lot of publicity
- Con: Cloudier winter than expected





Things That Renewables Owners Would Change

- Location
- Equipment
 - Type
 - Efficiency
 - Manufacturer
- Sooner!



Other Items That May Concern You

- #1 complaint was that the process took longer than expected
- Except for Fresh Hair, renewables have been completely transparent to production operations
- Everyone said they'd do it again if they had it to do over



Note to Assistance Providers...

- Many companies pursued on-site renewables as a result of direct outreach!
 - Fresh Hair
 - Bixby
 - Cordis
 - Varian Semiconductor



Questions??

